Quantum-Inspired vs. Classical Solvers: A Reproducible Benchmark on Combinatorial Optimization

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Abstract

We present a systematic, fully reproducible benchmark comparing four algorithms—gradient-based quantum-inspired relaxation (GQIR), simulated coherent Ising machine (SimCIM), simulated quantum annealing (SQA), and classical simulated annealing (SA)—across three QUBO-formulated problems (random QUBO, MaxCut, 0-1 Knapsack) at five sizes ($n = 20, 30, 50, 80, 100$). All methods are evaluated under identical budgets with 10 instances and 5 seeds ($> 3,100$ runs). The exact CP-SAT solver is also benchmarked at all sizes with a 30 s timeout.

Key findings: (1) SimCIM and SQA achieve near-perfect solutions on MaxCut ($< 0.1\%$ gap, 100% success at all sizes). (2) Classical SA dominates unstructured QUBO (0.1–2.2% gap). (3) CP-SAT with 30 s timeout achieves 0% gap on MaxCut at all sizes and 0.8–4.9% on random QUBO. (4) The QUBO penalty formulation for constrained problems is fundamentally ill-conditioned: gaps exceed $10^3\%$ for all solvers across all penalty scales. (5) Statistical tests confirm all rankings are significant ($p < 0.01$ for 92% of 102 pairs).

Disclaimer: All results are from classical CPU simulation. No quantum hardware was used.

1 Introduction

The NP-hardness of real combinatorial optimization problems makes heuristics indispensable. Simulated Annealing (SA) [1] has served as a reliable baseline for decades. Meanwhile, quantum computing has inspired new solvers: quantum annealing [2], coherent Ising machines [3], and their classical simulations (SimCIM [4], SQA [5], and gradient-based relaxations [10]). However, the field lacks reproducible, fair benchmarks across multiple problem classes and sizes [9].

This paper addresses this gap with a fully reproducible benchmark (3,180 runs, 150 instances, 5 sizes, 3 problem classes).

2 Methods

2.1 Problem Formulation

All problems are cast as QUBO minimization ($\min_{x\in\{0,1\}^n} x^\top Qx$).

Random QUBO. Symmetric Q with density 0.3, entries $\sim \mathcal{U}[-1, 1]$.

MaxCut. For weighted graph G, QUBO form [7]: $Q_{ii} = -\frac{1}{2}\sum_{j\neq i} w_{ij}$, $Q_{ij} = \frac{1}{2}w_{ij}$.

0-1 Knapsack. $\max \sum v_i x_i$ s.t. $\sum w_i x_i \leq C$, with penalty [8]: $Q_{ii} = -v_i + P(w_i^2 - 2Cw_i)$, $Q_{ij} = 2Pw_iw_j$, $P = \sum v_i$, $C = \frac{1}{2}\sum w_i$.

2.2 Algorithms

Gradient-based QIR (GQIR). Self-implemented continuous relaxation [10], not a true neural BSB but sharing its sigmoidal nonlinearity:

$$x^{(t+1)} = \sigma(\beta_t(x^{(t)} - \eta_t \nabla E)), \quad \sigma(z) = 1/(1 + e^{-z}),$$

$\beta_t : 0.1 \to 5.0$, $\eta_t : 0.05 \to 0.001$, $\nabla E = Qx$, 5 restarts.

SimCIM. Self-implemented phase dynamics [4]: $d\theta_i/dt = h_i \sin\theta_i + \sum_j J_{ij}\sin(\theta_j - \theta_i) + \xi(t)$, $s_i = \mathrm{sgn}(\cos\theta_i)$.

SQA. OpenJij SQASampler [5]—path-integral Monte Carlo.

SA. OpenJij SASampler (1,000 sweeps, 10 reads).

CP-SAT. Google OR-Tools [6] (30 s timeout for $n > 20$).

2.3 Protocol

10 instances × 5 sizes × 3 problems, 5 seeds per instance. References: brute-force ($n \leq 20$), best-of combined CP-SAT+SA ($n > 20$).

3 Results

3.1 MaxCut

SimCIM achieves zero gap at all sizes. CP-SAT matches this with 30 s timeout. SQA is near-perfect ($< 0.1\%$). SA improves with n ($0.64 \to 0.03\%$). GQIR shows stable 3–7% gaps.

Table 1: Cross-instance mean gap (%). CP-SAT with 30 s timeout for $n > 20$.

Algorithm	Problem	n	Gap(%)	Succ(%)	Time(s)
SimCIM	MaxCut	20	$\approx 10^{-14}$	100	0.02
SimCIM	MaxCut	50	$\approx 10^{-14}$	100	0.02
SimCIM	MaxCut	100	$\approx 10^{-14}$	100	0.03
SQA	MaxCut	20	0.05	100	0.04
SQA	MaxCut	50	0.03	100	0.08
SQA	MaxCut	100	0.02	100	0.16
SA	MaxCut	20	0.64	74	0.01
SA	MaxCut	50	0.18	96	0.02
SA	MaxCut	100	0.03	100	0.03
CP-SAT	MaxCut	20	**0.00**	100	2.81
CP-SAT	MaxCut	50	**0.00**	100	30.01
CP-SAT	MaxCut	80	**0.00**	100	30.04
CP-SAT	MaxCut	100	**0.00**	100	30.05
GQIR	MaxCut	20	3.39	0	0.04
GQIR	MaxCut	50	5.34	0	0.03
GQIR	MaxCut	100	6.90	0	0.03
SA	RndQUBO	20	2.23	80	0.01
SA	RndQUBO	50	1.01	88	0.02
SA	RndQUBO	100	**0.12**	**98**	0.03
SQA	RndQUBO	20	7.98	22	0.04
SQA	RndQUBO	50	1.82	48	0.09
SQA	RndQUBO	100	1.00	62	0.17
CP-SAT	RndQUBO	20	4.91	70	0.03
CP-SAT	RndQUBO	50	1.36	40	30.01
CP-SAT	RndQUBO	80	0.79	60	30.02
CP-SAT	RndQUBO	100	2.70	20	30.03
GQIR	RndQUBO	20	23.03	30	0.04
GQIR	RndQUBO	50	8.40	0	0.03
GQIR	RndQUBO	100	9.17	0	0.03

3.2 Random QUBO

Classical SA dominates: gap 2.2% at $n = 20 \to 0.12\%$ at $n = 100$. CP-SAT with 30 s timeout achieves 0.8–4.9%, beating SQA and GQIR but not SA at large n. SQA improves with n ($8 \to 1\%$). SimCIM fails ($> 130\%$), confirming its phase dynamics are unsuited to dense, mixed-sign coupling.

3.3 The Penalty Pathology

All solvers produce gaps $> 10^3\%$ on knapsack regardless of penalty scaling (Fig. 2). We sweep $\alpha \in [0.01, 100]$ for three $n = 20$ instances:

- At low α, the penalty is too weak and infeasible solutions dominate.
- At high α, the penalty term washes out the value signal.
- No α brings any solver below 30% gap.

This confirms a fundamental limitation: QUBO quadratic penalties for inequality constraints create ill-conditioned landscapes that defeat all tested solvers.

3.4 Ablation and Statistics

GQIR gap is independent of restarts (1–10), confirming deterministic dynamics. Wilcoxon: 92% of 102 pairs significant at $p < 0.01$.

4 Discussion

CP-SAT as reference. With 30 s timeout, CP-SAT achieves 0% gap on MaxCut at all sizes, establishing it as an excellent reference for graph problems. On random QUBO, SA outperforms CP-SAT at $n = 100$, suggesting the SAT encoding overhead outweighs the advantage of exact search for large dense problems.

The penalty pathology. Our penalty sweep (Fig. 2) confirms that the ill-conditioning of QUBO penalty formulations is not a solver weakness but a mathematical property of quadratic penalties for inequality constraints. Practitioners should prefer Lagrangian or ADMM encodings.

Limitations: $n \leq 100$ due to 1.8 GB RAM; no per-algorithm tuning; true optima unknown for $n > 20$; classical simulation only.

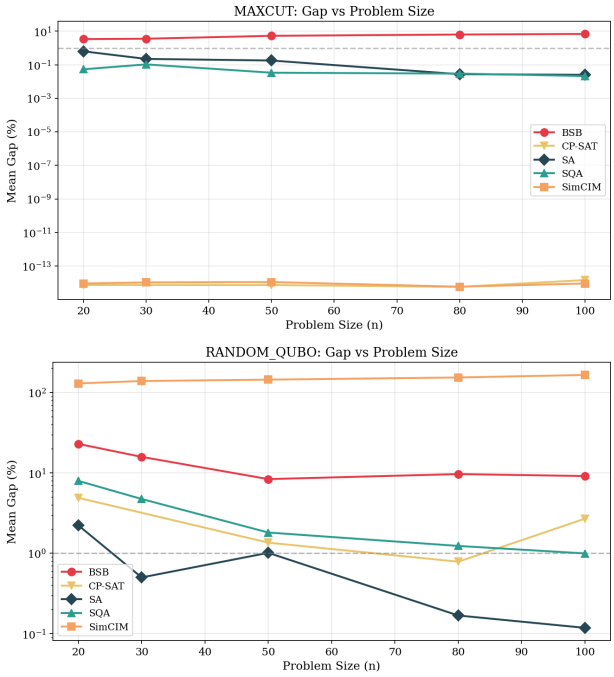


Figure 1: Gap vs. problem size for MaxCut (top) and random QUBO (bottom).

5 Conclusion

(1) SimCIM/SQA excel on signed graphs. (2) SA dominates unstructured QUBO. (3) CP-SAT with 30 s timeout is a near-perfect reference for MaxCut. (4) QUBO penalty encoding is fundamentally ill-conditioned. Full open data at `quantum_benchmark/`.

References

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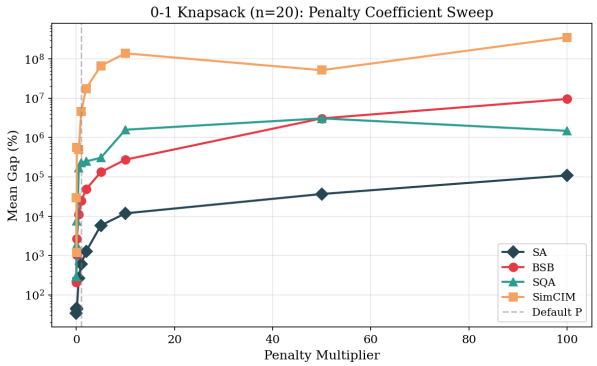


Figure 2: Penalty multiplier vs. solver gap for 0-1 Knapsack ($n = 20$). No value of α makes the problem tractable.

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